



## There is also another pathway other than the spinothalamic (important to localize and

characterize the source of the pain) and the spinoreticular (important to alert the brain about the painful stimulus). We have the sinomesencephalic, that is actually getting information from very deep structure that is called periaqueductal gray matter. The mesencephalo has this role of activating and regulating all the body structure, so together with the spinoreticular elements, this alertness is also reflected in several changes at the peripheral part of the body (heart rate increasing, sympathetic system increased). But more importantly this periaqueductal gray matter is a sort of an hub that modulates the incoming and outgoing painful information that goes to the brain, a gate that modulates the access of painful stimulation. Then there are two other pathways that are primarily related to two systems getting to cervical components that may actually send information from the cervical and the face and the neck up to the brain. One feature of these elements is that they relay to the reticular formation, but then they start to get other branches of this information, to the amygdala and the hypothalamus, but also the insula (primarily considered as the central pain area) and the cingulate cortex, and this final is important because it seems to be fundamental to monitor conflict or painful situation. --- explain the five main pathway

The five main pain pathways form an integrated network that processes different aspects of nociceptive information, from precise localization to emotional responses and autonomic reactions. <sup>[1]</sup> <sup>[2]</sup> <sup>[3]</sup>

### Spinothalamic Pathway

The spinothalamic tract is primarily responsible for localizing and characterizing the source of pain. It has two components: the neospinothalamic pathway carries sharp, well-localized "fast" pain with somatotopic organization allowing accurate identification of where the painful stimulus originates, while the paleospinothalamic pathway transmits dull, aching "slow" pain. Fibers from laminae I and V of the dorsal horn decussate and ascend to terminate in the ventral posterolateral (VPL) nucleus of the thalamus, then project to the primary somatosensory cortex for conscious pain perception. <sup>[3]</sup> <sup>[1]</sup>

### Spinoreticular Pathway

This pathway is crucial for alerting the brain about painful stimuli and initiating arousal responses. Neurons from the deeper dorsal horn layers project bilaterally to the reticular formation in the brainstem, then to intralaminar thalamic nuclei, which have widespread cortical projections. This diffuse connectivity mediates the alertness and autonomic responses to pain, including increased heart rate and sympathetic activation. <sup>[4]</sup> <sup>[5]</sup> <sup>[1]</sup>

## Spinomesencephalic Pathway

The spinomesencephalic tract originates in laminae I and IV-VI of the spinal dorsal horn, decussates, and ascends to multiple midbrain structures including the periaqueductal gray (PAG), cuneiform nucleus, and superior colliculus. The PAG serves as a critical hub that modulates both ascending painful information going to the brain and descending pain control signals projecting back to the spinal cord dorsal horn, functioning as a gate mechanism. This pathway is essential for pain modulation and the descending inhibitory control system. [2] [6] [4]

## Spinohypothalamic Pathway

This is a bilateral projection from the deep dorsal horn to both medial and lateral hypothalamus, with additional projections to the nucleus accumbens and septal nuclei. It plays a key role in the motivational and affective aspects of pain, integrating nociceptive information with autonomic and neuroendocrine responses. The hypothalamic connections mediate visceral and emotional responses to pain stimuli. [5] [1]

## Trigeminothalamic Pathways

Two pathways carry information from cervical, facial, and neck regions to higher brain centers. After relaying to the reticular formation, these pathways send branches to the amygdala and hypothalamus (mediating emotional and autonomic responses), the insula (considered the central pain area integrating sensory input with cognitive components), and the anterior cingulate cortex, which monitors conflict and painful situations and adds emotional valence to pain experiences. [7] [3] [5]

✱✱

1. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7859971/>
2. [https://taylorandfrancis.com/knowledge/Medicine\\_and\\_healthcare/Anatomy/Spinomesencephalic\\_tract/](https://taylorandfrancis.com/knowledge/Medicine_and_healthcare/Anatomy/Spinomesencephalic_tract/)
3. <https://nba.uth.tmc.edu/neuroscience/m/s2/chapter04.html>
4. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7355617/>
5. <https://nba.uth.tmc.edu/neuroscience/m/s2/chapter07.html>
6. [https://en.wikipedia.org/wiki/Periaqueductal\\_gray](https://en.wikipedia.org/wiki/Periaqueductal_gray)
7. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7224326/>
8. [https://www.wikidoc.org/index.php/Periaqueductal\\_gray](https://www.wikidoc.org/index.php/Periaqueductal_gray)
9. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6121522/>
10. <https://www.ncbi.nlm.nih.gov/books/NBK554391/>
11. <https://www.youtube.com/watch?v=xkMP4eXp1Oc>
12. [https://www.physio-pedia.com/Spinohypothalamic\\_tract](https://www.physio-pedia.com/Spinohypothalamic_tract)
13. [https://en.wikipedia.org/wiki/Spinotectal\\_tract](https://en.wikipedia.org/wiki/Spinotectal_tract)
14. [https://en.wikipedia.org/wiki/Spinomesencephalic\\_pathway](https://en.wikipedia.org/wiki/Spinomesencephalic_pathway)
15. <https://www.sciencedirect.com/science/article/pii/S0304395991901873>

